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Lessons Learned from First Application of Polymer Dispersed System (PDS) Chemical Water Shut-Off Technology in Carbonate Reservoir: A Case Study for Water Production Management in Mature Oil Field, UAE

Mohamed Mahmoud Abdelrahman, Saloua Tiar, Marinus Grutters, Amna Aldhanhani, Karma Alshammari, Ahmed Al Eissae, Nidhal Aljneibi, Aamna Al Tenaiji, and Mohamed Al-Attar, ADNOC Onshore, Abu Dhabi, United Arab Emirates; Sergei Matveev, Idzhat, Kazan, Russia

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Abstract

Excessive water production in oil reservoirs is a common challenge in mature fields, reducing oil recovery efficiency and increasing operational costs. Chemical Water Shut-Off (CWSO) techniques have emerged as a promising solution to control unwanted water production, thus enhancing oil recovery and extending the productive life of reservoirs.

Through a detailed case study, laboratory experiments, and a comprehensive workflow, the key selection criteria was investigated for applying Chemical Water Shut-Off (CWSO) treatment within heterogeneous carbonate formations.

This examination focused on the use of advanced Polymer Dispersed System (PDS) technology, which offers enhanced selectivity and efficiency in controlling unwanted water production. The workflow guided the diagnostic evaluation of water production sources, the design of tailored treatment plans, and precise chemical placement techniques. Overall, this study will demonstrate the application and evaluation of the first CWSO treatments using the PDS technology in one of the mature oil fields in UAE.

This new technology was applied in one oil producing well in Onshore, UAE where the well was producing with more than 90 % W.C and low oil production. Reservoir characterization, PLT and other diagnostic tools were utilized to design the job using new chemical technology. The pumping schedule and chemical amount was optimized during the job. However, a deviation from the planned injection program was necessary due to concerns for increasing bottomhole pressure, likely the result from different reservoir properties than expected from data analysis. Comprehensive pre and post job monitoring was utilized for assessing the treatment's effectiveness. This monitoring includes tracking reservoir response, production changes, and the movement of fluids within the reservoir. The analysis and lessons learned are presented to optimize the planning of future water shut-off jobs in the same reservoir.

Our findings demonstrated that the success of CWSO is highly dependent on proper diagnostic evaluation, precise chemical placement, and pre/after job monitoring. When executed successfully, CWSO

can significantly enhance reservoir performance by reducing water production, improving oil recovery, lowering operational costs, and extending the productive life of oil fields.

Introduction

Excessive water production in oil reservoirs is a widespread and persistent challenge, particularly in mature fields nearing the end of their productive life. Globally, water production exceeds oil production by more than three times, with over 210 million barrels of water being produced daily alongside hydrocarbons (Veil, 2015). This excessive water handling not only reduces oil recovery efficiency but also significantly increases operational costs due to the requirements for water separation, treatment, and disposal. In many fields, water handling can account for more than 50% of operating expenditures (Lyu et al., 2020). High water cuts defined as the ratio of water produced to total liquids can lead to the premature abandonment of otherwise productive wells, loss of recoverable reserves, and damage to both surface and downhole equipment due to scaling, corrosion, and erosion (El-Karsani et al., 2014).

To address these persistent problems, Chemical Water Shut-Off (CWSO) techniques have emerged as a cost-effective and minimally invasive solution. CWSO methods involve the injection of specialized chemical formulations such as polymer gels, cross-linked polymer systems, in-situ precipitates, or particle-based systems into the reservoir to selectively block high-permeability water channels (so-called "thief zones") while preserving or enhancing hydrocarbon flow from lower-permeability oil-rich zones (Al-Muntasheri et al., 2010). These treatments are particularly valuable in heterogeneous reservoirs where traditional mechanical shut-off solutions, such as cement plugs or mechanical packers, may be ineffective due to the complex nature of the reservoir's geology and fluid flow patterns (Seright, 1995).

Polymer-based gels, especially those using polyacrylamide and chromium cross-linkers, have demonstrated efficacy in reducing water production and improving sweep efficiency by diverting injected fluids toward unswept oil zones (Seright, 2006). Furthermore, recent advancements in nanoparticle and intelligent gel systems have enhanced the selectivity and long-term stability of these treatments under high-temperature and high-salinity conditions (Tang et al., 2022). By effectively targeting the sources of unwanted water production, CWSO techniques can prolong the economic life of wells, reduce water disposal volumes, and enhance overall recovery factors in aging fields.

Industry Challenge

High water production in aging carbonate reservoirs remains a significant operational and financial burden for oil and gas companies, particularly across the Middle East. In many mature fields, late-life wells routinely experience water cuts exceeding 85%, with some nearing 95% (Al-Hashmi et al., SPE-205432). This excessive water production not only impairs hydrocarbon recovery efficiency but also escalates costs associated with fluid lifting, separation, treatment, reinjection, or disposal. Water handling may account for up to 50–70% of the total operating expenses in such fields (Lyu et al., 2020; Veil, 2015).

Traditional Water Shut-Off (WSO) techniques, including mechanical methods such as bridge plugs and cement squeezes, have had limited success in these environments. Similarly, polymer-based chemical WSO systems, including crosslinked polyacrylamide gels, degrade under high-temperature conditions (above 200°F / 93°C), lose their mechanical integrity, and can fail to maintain conformance in fractured or vugular reservoirs (Seright, 2006; Al-Muntasheri et al., 2010). These limitations are further exacerbated by extreme reservoir heterogeneity typical of carbonate systems, which include a complex network of natural fractures, vugs, and a low-permeability matrix. Additionally, the economic burden of intervention is substantial.

Technology Gap

In response to the limitations of conventional WSO techniques, Polymer Dispersed Systems (PDS) have gained attention as an alternative method for managing unwanted water production in complex carbonate reservoirs. PDS technologies involve the injection of particle-laden or emulsion-based formulations

designed to selectively block high-permeability water pathways while minimizing oil flow impairment. In fields like Zabegalovskoe in Russia, the application of PDS has successfully reduced water cuts by 14–16% and improved oil production rates (Matveev et al., SPE-202835-MS). These results suggest potential for broader deployment of PDS in other challenging reservoir settings.

However, multiple technology gaps must be addressed before widespread application in Middle Eastern carbonate reservoirs is feasible. First, current PDS formulations have not been validated under the ultra-high salinity conditions (>200,000 ppm TDS) and elevated temperatures (>120°C) characteristic of many UAE reservoirs (El-Karsani et al., 2014). These extreme conditions challenge the chemical stability and particle dispersion of most known PDS systems, which may lead to premature agglomeration or ineffective placement.

Second, there is a lack of empirical data on the performance of PDS in fracture-dominated systems, which are prevalent in UAE fields due to tectonic activity and diagenetic processes. Fractures provide rapid pathways for water breakthrough, and the success of WSO agents in such environments depends heavily on the ability to propagate and plug selectively without bypassing the treatment zone (Seright, 1995; Al-Muntasheri et al., 2010).

Finally, existing PDS chemistries have not been optimized for microporous carbonate matrices, where pore throat sizes are in the order of microns or less. In such systems, the retention and injectivity behavior of dispersed polymer systems remains poorly understood, limiting their effectiveness for profile modification and conformance control.

Addressing these gaps requires new formulation strategies, enhanced experimental evaluation under representative reservoir conditions, and field-scale pilot trials tailored to the geological and operational realities of Middle Eastern carbonate fields.

Study Objectives

This study presents the first field trial of Polymer Dispersed System (PDS) technology in a mature carbonate reservoir in the United Arab Emirates. The primary objective is to assess the feasibility of using PDS for selective water shut-off in highly heterogeneous formations typical of the region. Specifically, the research addresses three critical aspects. First, it examines whether PDS formulations can effectively reduce water production in fractured and vugular carbonate systems under the extreme salinity and high-temperature conditions characteristic of UAE reservoirs. Second, it investigates how treatment protocols such as injection cycles, soak times, and post-treatment flush procedures must be adapted from those previously used in Russian carbonate fields to suit the geological and operational constraints of Middle Eastern reservoirs. Third, the study evaluates the comparative economic benefits of PDS treatments versus conventional water shut-off methods. Finally, the findings provide insights into how the treatment design can be improved for future job executions, including optimizing chemical concentrations, enhancing placement strategies, and refining operational workflows to ensure greater conformance control and long-term treatment stability.

Novel Contributions

This work contributes several important advancements to the field of chemical conformance control. First, it introduces a new PDS formulation that remains chemically and physically stable at temperatures up to 120°C and salinity levels over 225,000 ppm conditions previously thought to be unsuitable for polymer-based systems. Second, the study develops a bimodal particle distribution strategy within the PDS, enabling it to address both fracture and matrix conformance challenges by targeting different flow pathways simultaneously.

Significance to Industry

The implications of this study are both wide-reaching and highly relevant to regional production strategies. High water cuts often exceeding 80 % in late-life wells pose serious technical and economic challenges in these reservoirs, particularly due to their complex geology and harsh downhole conditions (Al-Hashmi et al., SPE-205432; Lyu et al., 2020). The findings of this study offer operators a technically sound and economically attractive alternative to conventional water shut-off solutions. By demonstrating the adaptation and possible modifications of Polymer Dispersed System (PDS) technology to high-salinity (>200,000 ppm) and high-temperature (>120°C) environments, the work significantly extends the applicability of chemical conformance methods beyond the more temperate settings where PDS has traditionally been used (Matveev et al., SPE-202835-MS; El-Karsani et al., 2014).

Polymer Dispersed System (PDS) Mechanism and Laboratory Validation

Chemical Composition & Mechanism of Action

The Polymer Dispersed System (PDS) formulation consists of two synergistic components specifically designed for the harsh conditions of Middle Eastern carbonate reservoirs, including high salinity (>200,000 ppm), elevated temperatures (>120°C), and complex pore structures.

The first component is a thermally stable polymeric dispersion made of fine polymer particles typically sulfonated or acrylamide-based suspended in a brine-tolerant fluid. This component targets high-permeability water channels (e.g., fractures, vugs) and remains stable under harsh ionic and thermal conditions (El-Karsani et al., 2014; Lyu et al., 2020).

The second component is dispersed particles that are essential for maintaining the stability and performance of slurries, especially in challenging environments like Middle Eastern brines. Their main functional importance lies in preventing particle aggregation, which helps keep the slurry uniform and free flowing. They also preserve particle integrity, ensuring that the physical and chemical properties of the particles remain consistent throughout use. This is particularly critical in high-salinity conditions, where maintaining dispersion is key to effective processing and reliable performance.

Together, these components form a selective water-blocking barrier that maintains oil flow while withstanding the reservoir's harsh environment. The system addresses limitations of conventional gels, offering improved thermal stability, conformance control, and adaptability for deployment in UAE and GCC carbonate fields (Matveev et al., SPE-202835-MS).

Sulfonated polyacrylamide (SPAM) copolymer specifically engineered for high-temperature, high-salinity (HTHS) reservoir conditions:

Table 1—Sulfonated polyacrylamide (SPAM) Specifications

Property	Specification	Significance for PDS
Molecular Weight	10-12 million Daltons	Provides optimal viscosity for particle transport
Sulfonation Degree	25-30%	Enables stability in 250,000 ppm TDS brines
Thermal Stability	Up to 130°C (266°F)	Maintains performance in Middle East reservoirs
Hydrolysis Resistance	<5% viscosity loss after 30 days at 120°C	Ensures long-term stability in PDS applications

Table 2—Dispersed Particles Characteristics

Characteristic	Requirement	Functional Importance
Surface Charge	Zeta potential: -30 to -50 mV	Prevents aggregation in slurry
Functional Groups	-OH, -SO ₃ ⁻	Binds polymers (e.g., SPAM)
pH Stability	5-9 (reservoir conditions)	Maintains particle integrity
Ionic Sensitivity	Stable at >200,000 ppm TDS	Critical for Middle East brines

Mechanism of Action

The flocculation reaction in Polymer Dispersed Systems (PDS) is a multi-step physicochemical process in which dispersed polymer particles interact and aggregate to form larger flocs. These aggregates selectively block high-permeability, water-producing zones within the reservoir, reducing unwanted water flow while preserving oil permeability. This controlled flocculation enables effective conformance control by targeting fractures and thief zones, improving sweep efficiency and reservoir management (Fig. 1).

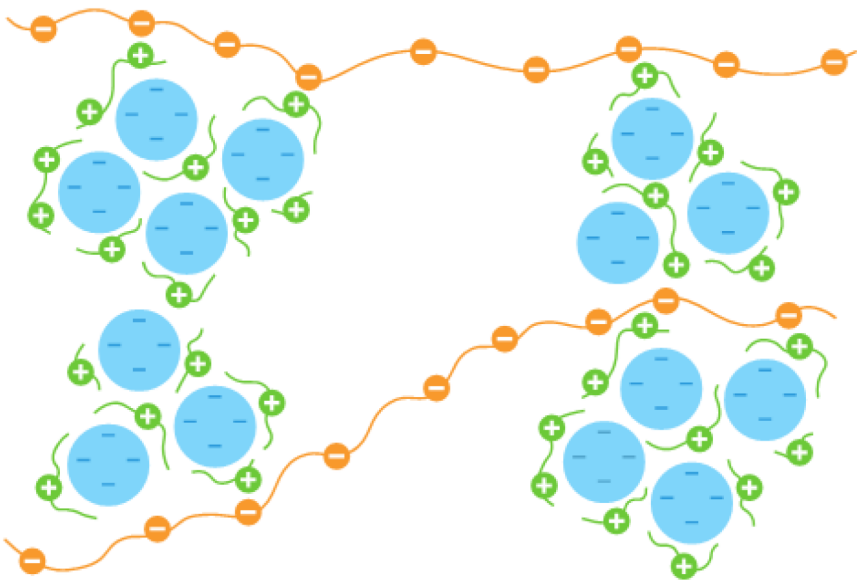


Figure 1—Bridging Flocculation

Below is a detailed technical breakdown:

Table 3—Bridging flocculation Stages and Mechanisms

Stage	Mechanism	Key Parameters
Polymer Adsorption	Polymer chains (e.g., PAM-X1) attach to particle surfaces via: - Electrostatic interactions (e.g., -SO₃⁻ on polymer ↔ Ca²⁺ on CaCO₃) - Hydrogen bonding (e.g., -OH groups) - Hydrophobic interactions	Polymer MW, particle zeta potential, salinity
Particle Bridging	Polymer chains extend between multiple particles, forming 3D networks (house-of-cards structure). This step is induced by the dispersed particles; the injection of the second batch that contains the dispersed particles reacts with the polymers of the first batch and starts forming the bridging.	Polymer concentration, shear rate
Aggregation	Bridged particles collide and form micro-flocs (1–100 μm) under reservoir flow conditions	Temperature, ionic strength

Stage	Mechanism	Key Parameters
Gelation	Micro-flocs merge into macro-flocs, creating a permeable mesh structure in water zones	pH, divalent cations ($\text{Ca}^{2+}/\text{Mg}^{2+}$)

Selective Activation in Water vs. Oil Zones

Table 4—Selective Activation in Water vs. Oil Zones

Zone	Behavior	Result
Water Zones	High salinity compresses double layer → particles aggregate via polymer bridges	RRF >30 (permeability reduction)
Oil Zones	Polymer remains soluble; particles disperse due to lack of ionic triggers	RRF <1.2 (no damage)

PDS Laboratory Data

Test Parameters:.

Table 5—Laboratory Validation Reference Tests

Condition	Lab Test
Temperature	250 F
Salinity (TDS)	225,000 ppm
Polymer Concentration	2,000 ppm sulfonated PAM
DP Concentration	1000 ppm
Core Permeability	2,100 mD (Bentheimer)

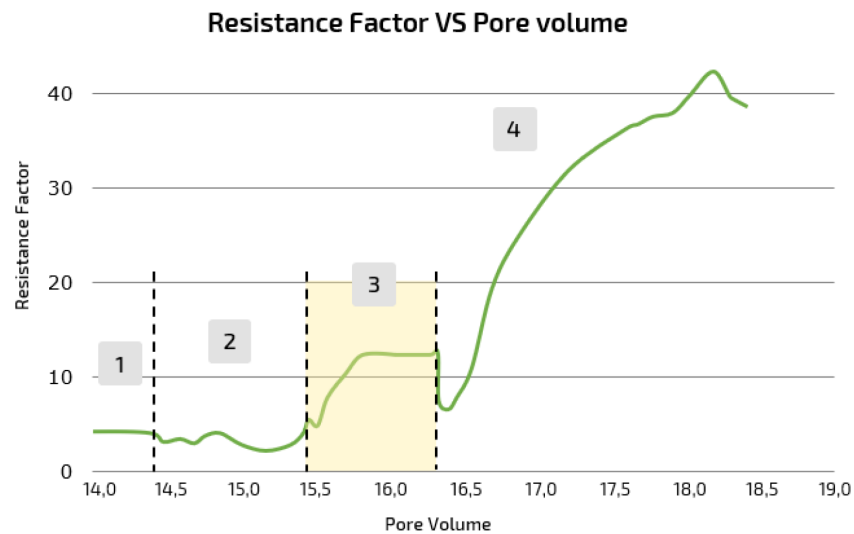


Figure 2—PDS Lab Test at Harsh Conditions (High Temperature, High Salinity)

Lab Results. Although a different core type was used in this lab test, the PDS lab test conducted under harsh conditions characterized by high temperature (250°F) and high salinity demonstrated excellent performance and stability. The polymer showed good propagation with a resistance factor (RF) of 4–5, indicating effective mobility control. Additionally, particles were successfully injected into the formation, confirming their compatibility and injectivity at elevated temperatures. Following a brine flush, the residual resistance factor

(RRF) reached 12, showing strong formation retention. Post polymer flush, the system stabilized at a significantly higher RF of 43, highlighting the enhanced performance and long-term effectiveness of the PDS formulation under extreme reservoir conditions.

Particle Selection Guide. Proper particle selection is essential for effective treatment based on reservoir permeability. For formations with permeability less than 10 millidarcies (mD) fine particles with 80% <1 μm are required, ensuring deep penetration and minimal formation damage. For formations with permeability greater than 10 mD larger particles in the 3–4 μm range can effectively block high-permeability zones while maintaining injectivity and performance.

Candidate Selection Criteria for Polymer Dispersed System (PDS) Treatments

The following section delineates the ideal candidates and favorable conditions for Polymer Dispersed System (PDS) Treatments, while also presents the characteristics of poor candidates. Additionally, it highlights the warning signs and challenges.

Reservoir Characteristics

Table 6—Reservoir Characteristics for Polymer Dispersed System (PDS) Candidates

Parameter	Ideal Candidate	Marginal Candidate	Poor Candidate
Permeability	5-500 mD (matrix)	<10 mD or >1000 mD	Unfractured <5 mD
Heterogeneity	Fractured/vuggy systems	Homogeneous sands	Chaotic thief zones
Temperature	<120°C (250°F)	120-150°C (modified PDS)	>150°C
Salinity	<100,000 ppm TDS	100,000-200,000 ppm	>250,000 ppm
Pore Throat Size	1-50 μm (match particle size)	<0.5 μm or >100 μm	Microporous carbonates

Well Completion & Production Data

Table 7—Reservoir Characteristics for Polymer Dispersed System (PDS) Candidates

Factor	Favorable Indicators	Warning Signs
Water Cut	50-90 %	<30% or >95%
Water Source	PLT-identified discrete zones	Diffuse water production
Production History	Sudden water breakthrough	Gradual water cut increase
Completion Type	Open hole/perforated	Gravel-packed
Nearby Injectors	>500m distance	<200m (pressure interference)

Fluid Properties

Table 8—Fluid Properties Suitable Conditions for Polymer Dispersed System (PDS) Candidates

Property	Suitable Conditions	Challenges
Oil Viscosity	<10 cP	>50 cP (mobility issues)
Oil API Gravity	>25°API	<15°API (asphaltene risk)
Water Chemistry	Ca^{2+} <5000 ppm	High $\text{Ca}^{2+}/\text{Mg}^{2+}$ (>10,000 ppm)

Well-A: PDS Treatment Proposal and Implementation

Well-A Overview

The candidate well is a horizontal open hole well exhibiting characteristics that make it a strong prospect for PDS treatment as from initial production, the well presented a high water cut of approximately 70%, suggesting early aquifer breakthrough or strong aquifer communication. Subsequent Production Logging Tool (PLT) runs confirmed that water production is primarily emanating from the toe of the horizontal section. This localized water influx makes the well an ideal candidate for PDS, as the treatment is aiming to selectively reduce water production from the toe while preserving or enhancing hydrocarbon recovery from the remaining productive zones.

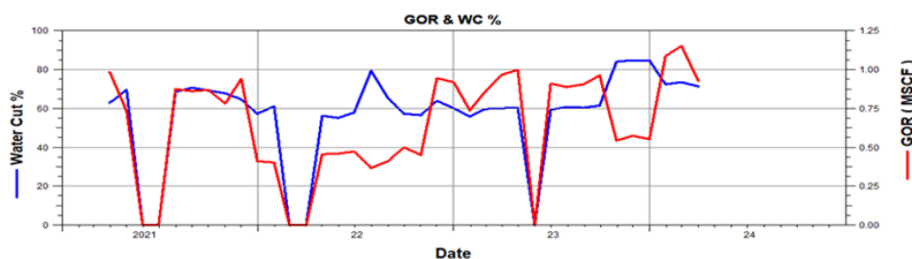


Figure 3—Well A performance plot

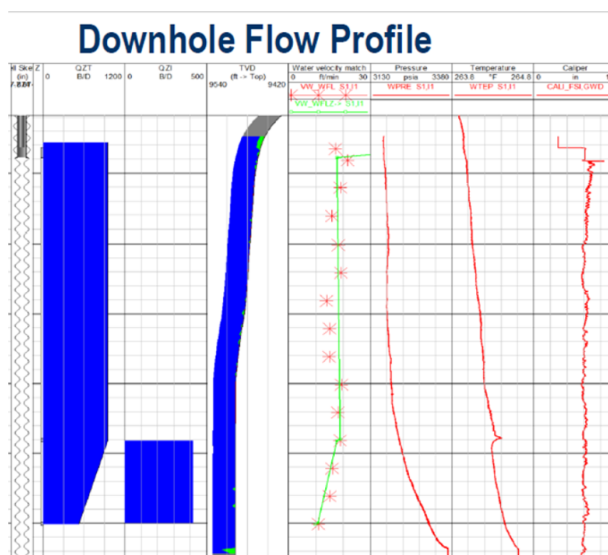


Figure 4—Well-A Latest PLT Results

PDS Treatment Parameters

Table 9—Polymer Dispersed System (PDS) Treatment Parameters

Component	Specification	Rationale
Polymer	Sulfonated PAM (0.1%, 10–12 MDa)	High-salinity (220,000 ppm) and thermal (200°F+) stability
Particles	3–4 μm	Targets fractures/vugs in lower zone
Injection Volume	3,000 bbl	Based on PLT identified water entry points
Shut-in Time	48 hours	Ensures PDS flocculation in water zones

Well-A PDS Treatment Summary

Table 10—Pre-Treatment Injectivity Test

Injection Rate (bbl/min)	Pressure (psi)	Notes
0.5	930	Baseline established
2.7	1,960	Max pressure below fracture gradient

CT Deployment Challenges. During the PDS treatment operation, tagging issues were encountered as multiple attempts to reach total depth (TD) were unsuccessful due to debris or wellbore restrictions. To overcome this challenge, a straight bar was added to the coiled tubing (CT) bottom hole assembly (BHA), which provided the necessary weight and rigidity to navigate the restrictions. Following this adjustment, TD was successfully tagged on the fourth attempt, enabling the continuation of the planned PDS deployment.

PDS Injection. The PDS treatment consisted of three alternating polymer/particle injection cycles, with each stage comprising 300 barrels. Following the treatment, post-job injectivity was measured at 1.5 barrels per minute with a surface pressure of 2,200 psi representing a 33% increase in pressure compared to pre-treatment conditions.

Table 11—Key Operational Data

Parameter	Value
Total Polymer Injected	300 bbl
Total Particles Injected	300 bbl
Fresh Water Used	1,113 bbl (spacer/cleanup)
Max WHP During Injection	2,000 psi
Shut-in Time	48 hours

Following the implementation of the job, well performance was meticulously monitored, and a comprehensive review was undertaken to identify the key challenges and lessons learned from the PDS treatment. These insights will be instrumental in guiding future job implementations.

Key Challenges & Lessons Learned from PDS Treatment in Well-A

Particle Size Optimization

Observation. Only 600 barrels of PDS chemicals were injected, which is merely 20% of the planned 3,000-barrel volume. The primary limiting factor was the high injection pressure, which reached 2,200 psi and prevented further placement. This elevated pressure was linked to the larger particles used during the design, which were intended to infiltrate the water in the toe via a high-permeability streak. However, due to the lower-than-expected permeability/porosity, the wrong dispersed particle size was selected. The increased pressure may have been caused by the caking of the open hole with the dispersed particle slurry, as they could not penetrate the reservoir. This likely limited the placement efficiency of the PDS treatment, particularly in areas dominated by micro-fractures or tight pore structures.

Recommendations:. For future treatments, it is advised to optimize the particle size distribution by incorporating smaller particles ($<1\ \mu\text{m}$) to improve penetration into matrix porosity and micro-fractures. Larger particles ($3\text{--}4\ \mu\text{m}$) should be retained selectively for targeting major fracture systems where higher

mechanical blockage is required. To ensure effectiveness, laboratory testing should be conducted to validate the performance of the revised particle size blend under representative reservoir conditions.

Planning and Operational Enhancement

Pre-Treatment Testing and Analysis.

Observation

Injectivity testing was performed during the job rather than beforehand, limiting the ability to evaluate well readiness and optimize treatment design. Additionally, there was insufficient integration of subsurface data, and wellbore conditions were not fully confirmed prior to testing.

Recommendations

It is recommended to conduct injectivity tests prior to the main treatment to properly assess the well's feasibility for chemical placement. These tests should be integrated with Production Logging Tool (PLT) data and openhole log analysis to improve reservoir characterization and optimize the PDS treatment design. Additionally, ensuring wellbore stability by confirming the absence of gas and maintaining homogeneous fluid conditions before testing will help achieve more accurate and reliable injectivity results.

Defined Operational Thresholds.

Observation

No clear minimum injectivity criteria were defined before mobilization, increasing the risk of operational delays or premature termination. Formation evaluation via pressure fall-off was not systematically implemented.

Recommendations

It is important to establish and communicate a minimum injectivity rate requirement before job mobilization to provide a clear go/no-go decision point for proceeding with the treatment. Additionally, conducting a fall-off test after the injectivity test will help evaluate pressure decline and estimate formation permeability. The shut-in period for this test should be optimized to gather reliable data while minimizing any disruption to ongoing production.

Assurance of PDS Mechanism.

Observation

There was a concern that the Polymer-Dissolved System (PDS) mechanism might not have formed effectively if the polymer and the dispersed particles had not adequately interacted during the treatment process. This lack of interaction could significantly impact the success of the chemical placement and the efficacy of the water shut-off.

Recommendations

It is crucial to ensure that the correct pumping depth is maintained and that the pumping schedule is optimized. Additionally, verifying that the dispersed particles effectively penetrate the reservoir is essential for achieving a successful treatment. This will ensure that the polymer and particles interact properly, forming a robust Polymer-Dissolved System (PDS) mechanism.

Summary and Conclusion

An advanced Chemical Water Shut-Off (CWSO) technique was applied to mitigate excessive water production in a highly water-prone well located within a mature carbonate reservoir. The operation was carefully planned and executed, targeting unwanted water flow while preserving hydrocarbon production. Following the treatment, detailed post-job analysis was conducted to evaluate the effectiveness of the intervention and to extract operational insights.

As a summary of the main key lessons, the PDS treatment in Well-A provided valuable insights to guide future operations. Improved pressure management specifically shutting in offset injector wells prior

to treatment can effectively reduce reservoir pressure and enhance injectivity, enabling more efficient and thorough chemical placement. Additionally, optimizing particle size by incorporating smaller particles in the 1 μm range significantly improves penetration into the matrix and micro-fractures, thereby increasing overall treatment effectiveness. Equally important is the optimization of the pumping schedule and chemical volumes to ensure proper contact between the polymer and dispersed particles, which is critical for achieving the desired PDS effect. Together, these improvements will enhance treatment performance and maximize water shut-off success in future wells.

These lessons learned provide a foundation for improving future CWSO applications through better candidate selection, treatment design, and execution strategies. The case offers a valuable reference for optimizing the planning and implementation of CWSO operations in challenging reservoir conditions.

When successfully deployed, CWSO technologies can significantly reduce water production, enhance oil recovery, lower operational and water handling costs, and extend the productive life of wells. This underscores the critical role of chemical conformance control in the long-term management of mature oil fields, especially where mechanical shut-off methods are not viable.

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